SYSTEM REQUIREMENTS

ZCOS Knowledge Engine

Canonical requirements for galaxy-partitioned, source-grounded understanding

Field	Requirement value
Status	Locked for implementation
Authority	ZCOS Architecture Foundation
Scope	Topics, Knowledge Map, Sources, Lexicon, Curation, retrieval, and migration
Version date	August 7, 2026
Migration sources	Existing ZAR Object Memory, Knowledge Ingestion, Curation, Lexicon, and retrieval services

Governing rule: One ZCOS Knowledge Engine. Eight galaxy Knowledge partitions. One canonical knowledge authority. Every item retains origin, evidence, provenance, status, and originating galaxy. All Knowledge unifies access without collapsing partition identity.

1. Purpose and Authority

This document translates the locked ZCOS Architecture Foundation into an implementation contract for the ZCOS Knowledge Engine. It defines what Knowledge is, how it differs from Memory and other domains, how knowledge enters and evolves, and which controls must govern every read, write, compilation, correction, promotion, and cross-galaxy use.

The ZCOS Architecture Foundation is authoritative wherever this document conflicts with an older ZAR specification or repository behavior. The existing repository is migration evidence. Its object, graph, evidence, ingestion, lexicon, curation, retrieval, and audit mechanisms should be preserved only where they comply with this contract.

1.1 Required outcomes

- Unify the overlapping Object Memory graph and Knowledge Ingestion graph under one canonical Knowledge authority.
- Replace Memory-backed curation with curation over the actual canonical Knowledge store.

- Give each authenticated user one logically unified Knowledge system divided into eight galaxy partitions.
- Expose the locked surfaces: Topics, Knowledge Map, Sources, Lexicon, and Curation.
- Make every knowledge item traceable to evidence and visibly classifiable as UGC / Uploaded or Extracted / Compiled under Sources.
- Prevent candidate, conflicting, stale, rejected, superseded, or unauthorized material from being presented as current substantiated understanding.

2. Non-Negotiable Invariants

Invariant	Required meaning
One engine	ZCOS owns one Knowledge Engine. No galaxy, service, filesystem graph, vector store, or admin view becomes a competing knowledge authority.
Authenticated ownership	Every user-owned operation requires a verified unified Identity. Missing or fallback owners fail closed.
Galaxy origin	Every user-owned knowledge item originates in exactly one galaxy partition. Authorized use elsewhere does not change origin.
Canonical records	Knowledge objects, claims, source records, and relationships are authoritative. Search indexes, summaries, prompts, and embeddings are derived.
Evidence required	No item may be promoted as substantiated Knowledge without traceable support appropriate to its type and scope.
Origin classified	Every item identifies UGC / Uploaded or Extracted / Compiled under Sources. Origin is metadata, not a sixth surface.
Context preserved	Conflicting claims may coexist when scope, time, source, or definition differs. Curation must not flatten context to force false agreement.
State-safe retrieval	Only authorized and lifecycle-eligible material may influence ordinary responses as current Knowledge.
Versioned evolution	Corrections, merges, supersessions, and deprecations preserve the prior state, evidence, reason, actor, and timestamp.
Domain separation	Memory, Identity, Glow, Settings, Desk / Projects, Files, History, and ZCOS governance remain separate authorities.

3. Scope and Domain Boundaries

3.1 Canonical Knowledge content

Content type	Canonical meaning
Topic	A governed subject identifier used to organize and retrieve related Knowledge.
Concept	An idea, framework, category, definition, method, or abstraction understood by ZCOS.
Claim	A source-attributed assertion that may be supported, disputed, limited, or unresolved.
Fact	A claim promoted as substantiated within a defined scope because its evidence and authority requirements are satisfied.
Rule	A source-grounded requirement, constraint, policy, or governing instruction with a defined scope.
System	A structured description of a product, process, architecture, organization, or operating system.
Relationship	A source-backed connection between canonical objects, with direction, type, scope, and lifecycle.
Source record	The authoritative evidence and provenance record supporting or challenging Knowledge.
Lexicon sense	One contextual meaning of a word, phrase, abbreviation, symbol, or community term.

3.2 Items routed elsewhere

Content	Canonical owner
Personal experiences, decisions, relationships, and events retained for recall	Memory
Name, email, profile image, and Privy wallet address	Identity
Explicit interaction preferences and capability controls	Settings
Inferred personal patterns, adaptation, and growth	ZWAP! Discovery -> Glow

Content	Canonical owner
Tasks, open questions that require execution, project state, and work products	Desk / Projects
Original uploaded files and stored artifacts	ZENITH Logos -> Scholar -> Files
Conversation transcripts, messages, and system activity	Conversation History / History; eligible sources, not Knowledge by default
System identity, constitutional policies, safety rules, and tool contracts	ZCOS Core / system governance

Boundary rule: A source may produce routed records in several domains, but each record has one canonical owner. Cross-domain links never merge Memory, Knowledge, Projects, Files, or Identity into one graph authority.

4. Engine Architecture

4.1 Required layers

Layer	Responsibility	Authority status
Intake and routing	Receives eligible material, verifies owner and galaxy, classifies domain, and creates source-bound candidates.	Processing only
Source ledger	Stores origin class, source identity, evidence locators, provenance, rights, and compilation lineage.	Canonical Sources authority
Canonical Knowledge store	Stores typed objects, claims, facts, rules, systems, lexicon senses, lifecycle, versions, and partition ownership.	Canonical Knowledge authority
Knowledge Map	Stores first-class, evidence-backed relationships among canonical objects.	Canonical relationships
Topic registry	Maintains canonical topic identities, aliases, hierarchy, and partition membership.	Canonical organization
Curation engine	Evaluates duplicates, conflicts, confidence, currency, gaps, health, promotion, merges, and supersession.	Governed mutation

Layer	Responsibility	Authority status
Retrieval indexes	Provides semantic, keyword, topic, entity, temporal, relational, and source search.	Derived and rebuildable
Audit ledger	Records writes, promotions, reviews, merges, conflict resolutions, cross-galaxy access, exports, and deletion.	Authoritative audit evidence
User surfaces	Presents Topics, Knowledge Map, Sources, Lexicon, Curation, and All Knowledge.	View and control layer

4.2 Processing sequence

- Accept an eligible source or compilation request from an authenticated context.
- Resolve the unified user Identity and originating galaxy before extraction or persistence.
- Classify the material as Knowledge, Memory, Identity, Settings, Glow, Desk / Projects, Files, History, or system governance.
- Create or reuse the authoritative source record and assign UGC / Uploaded or Extracted / Compiled origin.
- Extract semantic units and candidate objects without presenting them as confirmed truth.
- Compare candidates with canonical objects, claims, senses, and relationships for equivalence, conflict, context, and time.
- Persist source records and candidates atomically, then run the required curation policy.
- Promote, merge, supersede, reject, or hold for review according to evidence and authorization.
- Update derived indexes only after canonical writes succeed.
- Retrieve only through ownership, partition, lifecycle, currency, relevance, and conflict gates.

5. Ownership and Galaxy Partitions

5.1 Partition set

Partition	Default owner surface
ZAR	ZAR Nexys
ZYNC	ZYNC Canvas
ZETA	ZETA Control
ZENO	ZENO Unite

Partition	Default owner surface
ZYLO	ZYLO Compass
ZWAP!	ZWAP! Discovery
ZENITH	ZENITH Logos
ZILLION	ZILLION Prosper

5.2 Partition rules

- Every user-owned record contains ownerUserId and galaxyId. Neither may be inferred from an unsafe fallback.
- The active galaxy reads and writes its own partition by default.
- Admin Access grants explicit read, contribution, promotion, or administrative authority across partitions.
- Authorization must be evaluated at request time and recorded in the audit ledger.
- Cross-galaxy retrieval preserves the original galaxy, source bindings, lifecycle, and confidence.
- Revocation prevents new access and invalidates derived caches or context assembled under the revoked grant.

5.3 All Knowledge

ZCOS Command Desk -> All Knowledge is an authorized projection across the eight partitions. It is not a ninth partition, not a copied aggregate store, and not permission to erase galaxy origin. Results must remain filterable and attributable by galaxy.

6. Canonical Knowledge Record Model

Field	Requirement
id	Stable, globally unique object identifier.
ownerUserId	Verified unified Identity owner for user-owned Knowledge.
galaxyId	Originating galaxy partition.
objectType	Topic, concept, claim, fact, rule, system, relationship target, source, or lexicon sense as applicable.
canonicalName	Current canonical label, with aliases stored separately.
summary / definition	Concise current understanding within the object's defined scope.

Field	Requirement
properties	Typed fields appropriate to the object type; unknown values remain unknown rather than invented.
topicIds / entityIds	Links to canonical topic and entity identifiers.
sourceBindings	One or more authoritative source ledger references with evidence locators.
originClass	UGC / Uploaded or Extracted / Compiled, represented under Sources.
scope	Domain, jurisdiction, project, community, population, time, or other boundary required to interpret the item.
lifecycleState	Candidate, supported, confirmed, disputed, superseded, deprecated, rejected, or historical.
confidence	Evidence-proportional confidence with an explainable basis; never a substitute for state.
currency	Current, review_due, potentially_outdated, historical, or unknown.
version	Immutable revision history with current canonical revision pointer.
createdAt / updatedAt / reviewedAt	Distinct timestamps for creation, change, and curation review.

6.1 Claim and fact discipline

- A Claim records what a source asserts; it does not become a Fact merely because extraction confidence is high.
- A Fact is a promoted Claim whose evidence, authority, scope, conflict, and currency requirements are satisfied.
- User confirmation can establish user intent, internal canon, or project truth, but cannot alone establish an external empirical fact.
- Statements that depend on time, jurisdiction, population, version, or context must carry those boundaries.
- Unknown, disputed, or insufficiently sourced assertions remain Claims and are represented honestly.

7. Topics

Topics are canonical organization records and user-facing projections over Knowledge. They group the same objects shown in the Knowledge Map and Sources; they do not create duplicate knowledge stores.

7.1 Topic requirements

- Stable topic ID, canonical label, aliases, description, parent and child topics, originating galaxy, and lifecycle state.
- Many-to-many links between topics and knowledge objects, claims, sources, and lexicon senses.
- Support for manual assignment and curation-approved extraction.
- Duplicate topic detection, aliasing, merge history, and redirect from retired topic IDs.
- Partition-aware browsing and an authorized All Knowledge topic projection.
- A topic name does not prove a claim, establish source authority, or replace object classification.

7.2 Topic hierarchy

- Topic hierarchy must support broader, narrower, and related-topic links without forcing every topic into one rigid tree.
- Cyclic parent relationships must be rejected; related-topic relationships may be non-hierarchical.
- Galaxy-specific topic meanings remain distinct when merging would erase necessary context.

8. Knowledge Map

The Knowledge Map is the object-and-relationship view of the canonical Knowledge store. It must expose what ZCOS understands, how items connect, which evidence supports each connection, and where uncertainty remains.

8.1 Relationship record

Field	Required behavior
id	Stable relationship identifier.
subjectId / objectId	Canonical endpoints; dangling endpoints are not allowed in active state.
predicate	Controlled relationship type with direction and documented meaning.
scope	Context in which the relationship is asserted.
sourceBindings	Evidence supporting or challenging the relationship.
confidence	Evidence-proportional confidence independent from endpoint confidence.
lifecycleState	Candidate, supported, confirmed, disputed, superseded, deprecated, rejected, or historical.
validFrom / validTo	Temporal validity when known.

Field	Required behavior
version	Revision history for predicate, endpoints, scope, evidence, and status.

8.2 Graph rules

- Relationships are first-class records, not untraceable strings embedded in object properties.
- Graph traversal enforces user ownership, partition authorization, lifecycle, and source visibility at every hop.
- Curation may propose links but may not silently promote weak associations as substantiated relationships.
- Removing or superseding a relationship updates graph indexes without deleting unrelated endpoint objects.
- The graph must distinguish contradicted_by, supersedes, derived_from, supports, and merely related_to.

9. Sources

9.1 Required origin classification

Origin class	Meaning	Examples
UGC / Uploaded	The user directly supplied the knowledge statement or source material.	Typed input, direct correction, uploaded document, imported note, user-authored specification.
Extracted / Compiled	ZCOS identified, structured, summarized, synthesized, or researched the item from authorized material.	Extracted claim, compiled comparison, research synthesis, inferred relationship, normalized definition.

Classification rule: Origin class records how the knowledge item entered ZCOS. It does not determine truth, ownership, confidence, or authority. An Extracted / Compiled item may cite UGC / Uploaded material in its lineage, and both classifications must remain visible without collapsing into one label.

9.2 Source record requirements

Source field	Required behavior
sourceId	Stable identifier for the source record.

Source field	Required behavior
ownerUserId	Owner when the source is user-owned.
sourceGalaxyId	Galaxy in which the source entered ZCOS.
originClass	UGC / Uploaded or Extracted / Compiled.
sourceType	Conversation, direct text, file, URL, publication, database, research result, system event, migration, or compilation.
artifactRef	Reference to the original artifact in its canonical owner, such as ZENITH Files; the Knowledge Engine does not duplicate file storage.
title / author / publisher	Human-readable identity and authority metadata when known.
createdAt / publishedAt / accessedAt	Separate creation, publication, and retrieval dates where applicable.
version / finality	Version, edition, draft, final, or unknown.
locator	Message, page, section, timestamp, record, or segment location.
evidenceExcerpt	Minimum excerpt or normalized evidence necessary to verify the binding, subject to rights and privacy.
contentHash	Deduplication and integrity support when available.
rights / retention	Access, reuse, privacy, and retention restrictions.
lineage	Upstream sources and transformation method for Extracted / Compiled items.

9.3 Evidence and provenance rules

- Every promoted item must be traceable to sufficient evidence for its type and scope.
- Repeated copies of the same upstream source count as one evidentiary lineage, not independent corroboration.
- Source authority, independence, relevance, directness, recency, and consistency are evaluated separately.
- A source may support one field and contradict another; evidence attaches at object, claim, property, or relationship level as needed.
- Compiled knowledge must preserve transformation lineage, source set, compilation method, and version.
- Removing access to a source invalidates unauthorized excerpts and triggers review of dependent knowledge without automatically erasing unrelated valid evidence.

10. Lexicon

Lexicon is the contextual meaning authority inside the Knowledge Engine. It resolves terminology before reasoning and preserves distinct senses rather than flattening words with multiple cultural, technical, historical, or personal meanings.

10.1 Lexicon sense record

- One entry represents one meaning of a term; distinct meanings remain separate linked senses.
- Each sense stores canonical form, variants, definition, contexts, domains, communities, examples, relationships, confidence, evidence, source bindings, version, and lifecycle.
- User-owned entries require ownerUserId and originating galaxyId.
- System reference lexicon may be read-only ZCOS Core infrastructure; it must not be confused with a user's galaxy-owned vocabulary.
- Candidate, ambiguous, deprecated, rejected, and superseded senses are handled explicitly during resolution.

10.2 Resolution order

- Use an explicitly selected or user-confirmed sense when it matches the current context.
- Prefer the active galaxy's authorized vocabulary for galaxy-specific terms.
- Use cross-galaxy user vocabulary only when Admin Access permits it.
- Use domain, community, nearby text, project, and conversation context to rank remaining senses.
- Return ambiguity when materially different senses remain plausible; do not silently choose a harmful or meaning-changing interpretation.
- Record unresolved terms as curation candidates only when persistence is authorized; ordinary resolution remains read-only.

10.3 Required evolution from the repository

- Preserve the existing multi-sense entry model, evidence, authorities, candidates, confirmation, rejection, deprecation, and merge operations.
- Add galaxy scope to user-owned entries and authorization-aware cross-galaxy resolution.
- Move durable user lexicon records out of the shared filesystem fallback into the canonical user-owned data architecture.
- Prevent global promotion of user vocabulary without an explicit governed action and evidence appropriate to that scope.

11. Curation

Curation governs the quality and evolution of the actual canonical Knowledge store. It must not curate Core Memory, Project Memory, or Scratchpad Memory as substitutes for Knowledge.

11.1 Curation responsibilities

- Detect duplicates, aliases, overlapping concepts, weak relationships, orphaned objects, missing context, missing evidence, and open questions.
- Detect contradictions while distinguishing true conflict from differences in time, scope, definition, source, or jurisdiction.
- Evaluate confidence, source independence, completeness, authority, currency, version, relationship density, and review history.
- Propose confirmation, expansion, contradiction, supersession, merge, replacement, creation, deprecation, or clarification.
- Generate concise questions only when the answer could materially change classification, storage, retrieval, or conflict resolution.
- Maintain living topic collections and source-backed connections without creating a second canonical store.

11.2 Knowledge health

Health dimension	Required interpretation
Completeness	Required typed fields and scope are present.
Evidence quality	Evidence is relevant, direct enough, authoritative enough, and traceable.
Source diversity	Independent upstream lineages support the item; duplicates do not inflate diversity.
Confidence	Calibrated to evidence and uncertainty, not extraction fluency.
Currency	The item remains valid for its time-sensitive scope.
Conflict	Material unresolved contradictions reduce ordinary retrieval authority.
Context depth	Meaning, scope, dependencies, and relationships are sufficient for reliable use.
Verification	Required user, domain, or source verification has occurred.

Health scores support triage; they do not directly determine truth. The contributing dimensions, penalties, evidence, and review result must remain inspectable.

11.3 Human confirmation boundaries

- Major promotions, merges, replacements, canonical renames, and conflict resolutions require confirmation when they materially change long-term user or project authority.
- Safe deterministic normalization may proceed automatically when meaning, ownership, scope, and evidence are unchanged.
- External factual claims still require evidence appropriate to the claim even when a user approves storing them.

12. Knowledge Lifecycle

State	Entry condition	Ordinary retrieval behavior
Candidate	Extracted, compiled, imported, or proposed but not sufficiently evaluated.	Excluded as current truth; available to Curation and clearly labeled review contexts.
Supported	Has relevant evidence but awaits required confirmation, additional authority, or final review.	May be used with attribution and uncertainty when appropriate; never presented as confirmed fact.
Confirmed	Meets the applicable evidence, authority, scope, conflict, and review requirements.	Eligible as current Knowledge within authorization and currency limits.
Disputed	Material evidence or an authorized correction challenges the item.	Excluded from silent truth use; retrieve competing claims and explain uncertainty when relevant.
Superseded	A newer canonical revision or item replaces it for current use.	Historical use only; current retrieval resolves to the successor.
Deprecated	No longer recommended or valid for ordinary use, but retained for history or compatibility.	Excluded from ordinary current retrieval unless the query concerns legacy state.
Rejected	Determined not to qualify or explicitly rejected by authorized review.	Never used as Knowledge; retained only as minimal review evidence where allowed.
Historical	Valid only for a prior period or historical interpretation.	Eligible only when temporal context makes it relevant and clearly labeled historical.

12.1 Transition rules

- Extraction confidence cannot directly move a Candidate to Confirmed.
- Promotion records the evidence, policy, actor, timestamp, and previous state.
- A correction creates a new revision; the prior revision remains attributable and non-current.
- A merge retains aliases, source bindings, prior IDs through redirects, and the complete merge audit.
- A dispute does not silently delete the challenged item or overwrite its sources.
- Curation re-evaluates currency according to the item's domain and scope rather than one universal expiration period.

13. Confidence, Confluence, and Currency

13.1 Confidence contract

- Confidence represents support for a scoped proposition, not general certainty about an entire object.
- The system must preserve separate source, extraction, claim, relationship, and overall confidence where each is material.
- Corroboration increases confidence only when evidence is independently derived and meaningfully relevant.
- A directly authoritative primary source may outweigh many derivative sources without requiring false numerical equivalence.
- Confidence changes retain the prior value, new value, evidence delta, method, and reason.

13.2 Currency contract

- Each time-sensitive item defines or inherits a review policy suitable to its domain and scope.
- Current, review_due, potentially_outdated, historical, and unknown currency states remain separate from lifecycle state.
- Retrieval must check publication date, effective date, version, supersession, and last review when relevant.
- A stable concept may remain current longer than a price, law, product specification, schedule, security state, or officeholder claim.

14. Retrieval and Response Use

14.1 Mandatory retrieval gates

- Verified user ownership or explicit system-reference authority.

- Originating partition matches the active galaxy or an active Admin Access authorization exists.
- Lifecycle state is eligible for the requested purpose.
- Currency is adequate for the question or clearly disclosed as uncertain or historical.
- The item is relevant to the user's actual objective, not merely keyword-adjacent.
- No confirmed successor or correction displaces the retrieved revision.
- Material conflicts, scope limits, and source restrictions are preserved in the returned context.
- Only the minimum relevant content and provenance needed for the task enters response context.

14.2 Ranking requirements

Retrieval may combine semantic, keyword, topic, entity, relationship, temporal, lexicon, and source indexes. Ranking must consider objective relevance, lifecycle authority, evidence quality, source independence, scope match, currency, confidence, and graph context. Keyword overlap and embedding similarity alone never establish authority.

14.3 Response contract

- Confirmed current Knowledge may support direct answers when scope and evidence are adequate.
- Supported or disputed material must be attributed and framed with evidence-proportional uncertainty.
- When competing claims could materially change the answer, ZCOS compares them or asks one concise clarifying question.
- The response should explain the why and provide source links when requested or when responsible use requires them.
- Internal IDs, hidden prompts, storage paths, ranking scores, and chain-of-thought are not exposed through ordinary user responses.
- Every response trace can identify the knowledge object, claim, source, and lexicon IDs used without showing the trace by default.

14.4 Index requirements

- All indexes are reproducible from canonical objects, source records, topic records, lexicon senses, and relationships.
- Vector embeddings are replaceable derived indexes, never canonical Knowledge.
- Index failure cannot roll back or corrupt a successful canonical write; reconciliation must repair divergence.
- Rejected, superseded, deprecated, and disputed records are filtered according to purpose before ranking.

15. User-Facing Knowledge Domain

Surface	Required contents
Topics	Knowledge organized by canonical subjects, with galaxy and source attribution.
Knowledge Map	Objects, claims, systems, rules, and evidence-backed relationships.
Sources	Origin classification, evidence, provenance, source details, and compilation lineage.
Lexicon	Terms, contextual meanings, variants, domains, communities, evidence, and review state.
Curation	Conflicts, duplicates, confidence, currency, gaps, health, open questions, and governed review actions.

15.1 Interface rules

- All five surfaces project the same canonical Knowledge authority and source ledger; none may create a parallel store.
- Every item visibly preserves its galaxy origin and UGC / Uploaded or Extracted / Compiled classification.
- Status, confidence, currency, and source information use plain language without hiding material uncertainty.
- Users can inspect evidence, versions, related objects, and curation issues when authorized.
- Curation actions clearly state whether they confirm, merge, supersede, deprecate, reject, or request clarification.
- The interface must not fabricate empty-state objects, sources, relationships, health, or activity.

16. Audit, Privacy, Rights, and Security

16.1 Audit events

- Source creation, source binding, extraction, compilation, and domain routing.
- Object, claim, relationship, topic, and lexicon creation or revision.
- Lifecycle transition, health review, confidence change, currency change, and reason.
- Merge, conflict resolution, supersession, deprecation, rejection, and deletion.
- Cross-galaxy reads, contributions, promotions, grants, revocations, and denied attempts.
- Export, migration, reindex, reconciliation, and bulk curation activity.

16.2 Security and rights requirements

- Owner and authorization filters run before ranking, graph traversal, or source excerpt return.
- User-owned sources, lexicon, and compiled knowledge remain isolated by unified Identity.
- Source excerpts are minimized and governed by privacy, retention, licensing, and access rights.
- Original files remain under ZENITH Files; Knowledge stores references, evidence, metadata, and derived records, not duplicate artifact storage.
- Production canonical data uses the durable database architecture. Repository files may be fixtures, exports, staging, or verified read-only migration sources only.
- No raw personal exports, uploaded documents, or runtime user Knowledge may be committed to Git.
- Deletion or access revocation cascades to derived excerpts, indexes, summaries, caches, and unauthorized cross-galaxy material.

17. Existing Systems: Preserve, Change, Retire

17.1 Preserve and evolve

Existing capability	Required evolution
Object Memory typed objects and relationships	Move reusable object infrastructure into the shared object framework; route Memory and non-Knowledge types to their true authorities.
Knowledge Ingestion source analysis and semantic units	Retain behind authenticated owner, galaxy, origin classification, source ledger, and candidate lifecycle requirements.
Knowledge Ingestion graph and conflict model	Migrate into the single canonical Knowledge store rather than remaining a shared filesystem authority.
Object source references and evidence	Normalize into canonical source bindings with stable IDs, locators, rights, lineage, owner, and galaxy.
Lexicon multi-sense model	Preserve and add user ownership, galaxy scope, durable storage, and authorization-aware resolution.
Curation health and issue detection	Run against canonical Knowledge objects, claims, relationships, topics, sources, and Lexicon instead of Memory classes.
Context inquiry and open questions	Retain only for material uncertainties that change classification, storage, retrieval, or conflict handling.

Existing capability	Required evolution
KnowledgeService orchestration	Separate Memory, system governance, personalization, project context, and Knowledge retrieval while preserving an ordered response context.
Selective retrieval	Replace keyword-only authority with multi-index ranking and mandatory ownership, partition, lifecycle, currency, scope, and conflict gates.

17.2 Retire as canonical authority

- The undifferentiated Object Memory graph as authority for projects, rules, tasks, preferences, profile, Memory, and Knowledge.
- The separate Knowledge Ingestion filesystem graph as a production canonical store.
- Curation reports built from Core Memory, Project Memory, and Scratchpad Memory instead of canonical Knowledge.
- Shared Lexicon filesystem storage as the durable home for user-owned vocabulary.
- Timeline and Decisions as Knowledge surfaces; they route to History and Memory / Projects according to their canonical roles.
- All retrieval instructions that call mixed Object Memory content ground truth without lifecycle, conflict, source, or authorization checks.

18. Migration Requirements

Phase	Required result
Inventory	Identify every object graph, knowledge table, source store, lexicon store, curation report, prompt builder, index, cache, route, export, and filesystem fallback.
Freeze authority	Declare the new canonical boundaries before migration so legacy writers cannot continue producing mixed truth.
Classify domain	Route each legacy object to Knowledge, Memory, Identity, Settings, Glow, Desk / Projects, Files, History, governance, quarantine, or rejection.
Resolve ownership	Require verified ownerUserId for user-owned material; quarantine rather than guess when ownership is absent.
Assign galaxy	Use verified source and project context; quarantine ambiguous records rather than inferring origin from labels alone.

Phase	Required result
Normalize sources	Create source ledger records, origin classifications, evidence locators, rights, hashes, and compilation lineage.
Normalize Knowledge	Create canonical objects, claims, relationships, topics, lexicon senses, lifecycle states, currency, and version history.
Reconcile graphs	Merge true duplicates across the two graphs while preserving aliases, IDs, conflicts, sources, and redirects.
Rebuild curation	Run health, conflict, duplicate, gap, confidence, and currency review against canonical Knowledge.
Reindex	Build semantic, keyword, topic, entity, temporal, relational, lexicon, and source indexes from canonical records.
Verify	Run ownership, partition, source, lifecycle, curation, Lexicon, retrieval, and cross-galaxy acceptance tests.
Cut over	Route all production reads and writes through the ZCOS Knowledge Engine; retain old stores only as approved read-only migration evidence until decommissioned.

18.1 Migration guardrails

- Do not delete either legacy graph until record counts, ownership, source lineage, conflicts, and canonical mappings are verified.
- Do not bulk-promote candidates merely because they existed in an old graph or appeared in prompts.
- Do not convert uploaded files into Knowledge without preserving the original artifact reference and UGC / Uploaded lineage.
- Do not move personal Memory, profile data, tasks, or project state into Knowledge to simplify migration.
- Migration reruns must be deterministic and idempotent or create an explicit reconciled revision.

19. Required Service Contracts

Exact route names and storage technology belong to the build specification. Regardless of interface shape, the engine must expose equivalent authenticated operations for:

- Create or resolve a source record and classify its origin.
- Analyze source material without persistence, and ingest approved material as candidates.
- Create, retrieve, revise, and version canonical objects, claims, facts, rules, systems, topics, and relationships.

- List Knowledge by Topic, Map relationship, Source, Lexicon sense, Curation status, Galaxy, and All Knowledge filters.
- Promote, dispute, merge, supersede, deprecate, reject, and delete with governed audit.
- Resolve terms, phrases, and full text using authorized Lexicon scopes.
- Register, confirm, reject, deprecate, merge, and relate Lexicon senses.
- Run curation review, evaluate incoming material, inspect health, and answer material open questions.
- Retrieve ranked response context with source, lifecycle, scope, currency, and conflict metadata.
- Grant and revoke cross-galaxy Knowledge access through Admin Access.
- Rebuild indexes, reconcile partial failures, export authorized records, and inspect audit events.

20. Acceptance Criteria

20.1 Ownership and isolation

- User A cannot create, list, retrieve, curate, export, or delete User B's Knowledge or user vocabulary.
- Missing, invalid, or fallback user IDs fail clearly and create no record.
- External channels without verified unified Identity cannot access user-owned Knowledge.

20.2 Partition behavior

- Knowledge created in ZYNC retains ZYNC origin and is available to ZYNC by default.
- ZAR cannot retrieve ZYNC Knowledge without the required Admin Access authorization.
- Authorized cross-galaxy use preserves source bindings, ZYNC origin, lifecycle, and audit evidence.
- Revocation blocks later access and invalidates derived cross-galaxy context.

20.3 Source behavior

- Every knowledge item visibly reports UGC / Uploaded or Extracted / Compiled under Sources.
- An Extracted / Compiled item retains the complete upstream source lineage and transformation record.
- Copied versions of one upstream source do not count as independent corroboration.
- Removing source access prevents unauthorized excerpt retrieval and flags dependent Knowledge for review.

20.4 Lifecycle and truth

- Candidate content never appears as confirmed current fact.

- A Claim cannot become a Fact solely through extraction confidence or user enthusiasm.
- A confirmed correction displaces the prior revision in ordinary retrieval while preserving history.
- Disputed and materially conflicting claims are not silently flattened or overwritten.
- Historical and deprecated items appear only when the query requires their context and remain labeled.

20.5 Topics, Map, and Lexicon

- Topics, Map, Sources, Lexicon, and Curation project the same canonical authority without duplicate object creation.
- Merged topics and objects preserve aliases, source bindings, version history, and stable redirects.
- Graph traversal cannot cross an unauthorized partition at any relationship hop.
- A term with distinct senses remains distinct and resolves by context or returns ambiguity.
- User vocabulary cannot become global system vocabulary without a governed promotion action.

20.6 Curation

- Curation evaluates canonical Knowledge rather than Core, Project, or Scratchpad Memory substitutes.
- Health results expose contributing dimensions and do not directly declare truth.
- Duplicate, conflict, gap, weak-evidence, stale, and orphaned findings link to the exact affected records.
- Major merges, replacements, and conflict resolutions enforce their confirmation policy.

20.7 Retrieval quality

- Retrieval returns only authorized, eligible, current-enough, scope-matched, objective-relevant records.
- Primary authority and independent evidence outrank repeated derivative copies.
- Vector index loss can be repaired from canonical records without losing Knowledge or source lineage.
- Response traces identify used objects, claims, sources, and Lexicon senses without exposing internal reasoning by default.

21. Implementation Guardrails

Do not build: a third knowledge graph, a shared user-data folder, automatic upload-to-truth promotion, a single flattened claim per topic, galaxy-blind Lexicon resolution, Memory-backed Knowledge curation, keyword-only truth retrieval, or fallback-owner behavior.

- Preserve the approved five-surface Knowledge structure and existing visual design unless a separate UI specification authorizes change.
- Integrate reusable object, source, graph, ingestion, Lexicon, curation, and retrieval components before creating replacements.
- Make the minimum safe migration changes required to establish one authority.
- Do not remove working logic or legacy stores until the replacement path passes migration and acceptance tests.
- Treat this document as the requirements authority for the Knowledge Engine build and the later ZCOS and ZAR specifications.

Appendix A. Source Authority

Source	Role in these requirements
ZCOS Architecture Foundation, August 7, 2026	Primary authority for the five Knowledge surfaces, galaxy partitions, All Knowledge, source-origin classification, object framework, and domain boundaries.
ZCOS Memory Engine Specification, August 7, 2026	Compatibility authority for shared object infrastructure, partition authorization, provenance, versioning, retrieval gates, and separation of Memory from Knowledge.
ZAR Specification, August 4, 2026	Migration evidence for Object Memory, Knowledge Ingestion, KnowledgeService, Context Inquiry, Knowledge Curation, Lexicon Authority, routes, stores, and known exceptions.
ZAR + USER Interaction Guidelines and Behavior	Governs evidence over indicators, confluence, context, probabilistic reasoning, state tracking, explainability, learning from outcomes, and system-integrity behavior.
ZedAI repository audit	Verifies overlapping graph authorities, current types and statuses, filesystem persistence, Memory-backed curation, keyword Object Memory retrieval, and Lexicon scope gaps described in migration sections.